

Online Resource 1: Interaction-Aware Benchmarking of Exact-Gradient Signal and Finite-Shot Resolvability in a Four-Qubit Hybrid Quantum Neural Network

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1 Scope and frozen provenance

This supplement reports the corrected centered H1–H4 family and prospectively planned post-primary packages frozen before manuscript revision. The machine-readable source of truth is `results/final_submission_v1/manifest.json`. The numerical tag is `submission-numerical-results-freeze-v1`; the numerical-results commit is `f2c5f496bd9223aa1aa0f00382fafaf1ab95bfc05`, and freeze provenance is `6ec255cab39a516bfc1cd188c71e0225713605c4`. Historical files remain archived under `results/superseded/`; they are not current primary results.

2 Factor-coding correction and estimands

The adopted factors are

$$E_c = E - \frac{1}{2}, \quad L_c = L - \frac{1}{2}, \quad R_c = R - \frac{1}{2},$$

with values $\{-1/2, +1/2\}$. Because $E_c L_c R_c$ remains in each applicable model, centered lower-order interactions average equally over the third binary factor. Direct-0/1 lower-order coefficients instead describe simple interactions at the third factor’s reference level. The parameterizations yield identical fitted values, likelihood, residuals, and random effects; only the interpretation of lower-order coefficients changes. The correction was an implementation–interpretation correction rather than result-dependent

Hyp.	Estimand	Estimate	SE	Wald 95% CI	Raw p	Holm p	Decision
H1	$E_c L_c$, exact	0.004043	0.001081	[0.001924, 0.006162]	0.000185	0.000739	Reject
H2	$E_c L_c$, end-to-end	0.014338	0.005145	[0.004255, 0.024422]	0.005321	0.015963	Reject
H3	$E_c R_c$, end-to-end	-0.011615	0.005144	[-0.021697, -0.001534]	0.023937	0.047875	Reject
H4	$L_c R_c \times \text{depth}$	-0.010179	0.005362	[-0.020688, 0.000331]	0.057658	0.057658	Do not reject

Table 1 Corrected primary family. Decisions follow model-based Wald tests with Holm adjustment. Intervals in the final column are initialization-cluster percentile bootstrap intervals. H1 used 2,000 completed fits with zero failures; H2–H4 used 443 completed nested-bootstrap fits. Iterations are distinct from the 50 independent initialization clusters. H2 and H3 are model-based rejections not corroborated by their bootstrap intervals; H4 remains unresolved.

D	Original estimate	Original 95% CI	Independent-seed estimate	Independent-seed 95% CI
1	-0.005251	[-0.013708, 0.003206]	0.001862	[-0.004255, 0.007379]
2	-0.003579	[-0.009560, 0.002401]	0.003120	[-0.003120, 0.009360]
3	0.012588	[0.007706, 0.017471]	0.006250	[0.001875, 0.010625]
4	0.004664	[0.000435, 0.008892]	0.007885	[0.003943, 0.011827]
6	0.003446	[-0.000007, 0.006898]	0.010870	[0.003446, 0.018300]
Equal depth	0.002374	[-0.000164, 0.004911]	0.005997	[0.001999, 0.010000]
Observation/parameter weighted	0.004043	[0.001929, 0.006157]	0.007726	[0.003402, 0.011923]

Table 2 Post-primary H1 depth contrasts and weighting summaries. Observation and matched-parameter weights are identical: 0.0625, 0.125, 0.1875, 0.25, and 0.375 at $D = 1, 2, 3, 4, 6$. Intervals are unadjusted Wald 95% intervals.

model selection. H1 and H2 use $E_c L_c$, H3 uses $E_c R_c$ averaged over L , and H4 uses $L_c R_c$ moderated by centered/scaled depth.

3 Corrected primary family and bootstrap

For H2–H4, outer resampling selected complete initialization clusters and retained all configurations, depths, budgets, and matched parameters; within selected scientific cells, finite-shot replicates were resampled and SNR was recomputed before refitting. H1 resampled complete initialization clusters. Percentile intervals are reported without treating either Wald or bootstrap inference as universally preferable.

4 Independent-seed H1 and depth weighting

The independent-seed centered H1 estimate was 0.007726 (Wald 95% CI [0.005592, 0.009860]); its 2,000-fit, zero-failure bootstrap interval was [0.003402, 0.011923]. The direction was retained but magnitude remained uncertain.

Original moderation tests were mixed-model $\chi^2(4) = 22.844$, $p = 0.000136$, and cluster-robust $\chi^2(4) = 6.008$, $p = 0.1985$. Independent-seed tests were $\chi^2(4) = 7.563$, $p = 0.1090$, and $\chi^2(4) = 1.810$, $p = 0.7706$. The original conclusion is covariance-sensitive; cross-seed magnitude differences have no clear depth localization under the frozen rule.

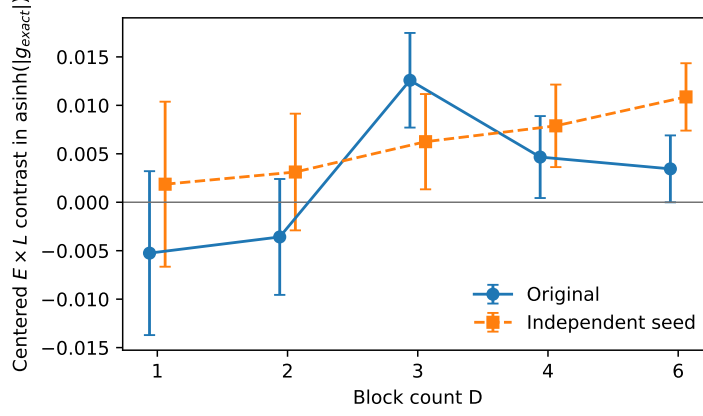


Fig. 1 Post-primary centered H1 contrasts by block count. Bars are unadjusted Wald 95% intervals; zero is the reference. Dataset identity is encoded by shape and line style, not color alone. No monotonic depth law is assumed.

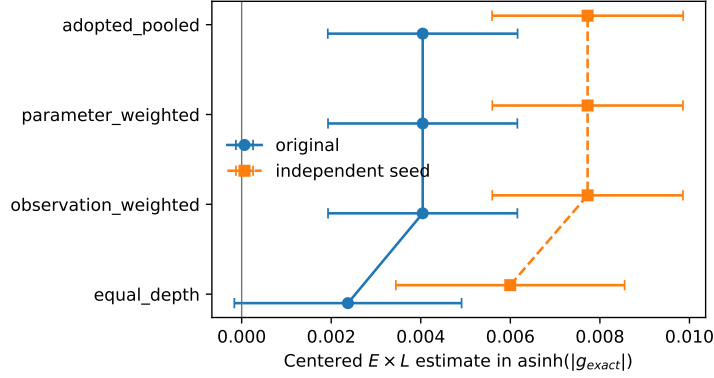


Fig. 2 Post-primary equal-depth, observation/matched-parameter, and adopted pooled H1 summaries. Bars are Wald 95% intervals. Weightings answer different questions and were not selected by significance.

5 H3 post-primary explanatory analyses

Depth-specific L-averaged estimates at $D = 1, 2, 3, 4, 6$ were 0.006642, 0.000696, -0.026492 , -0.011553 , and -0.011255 . Joint moderation did not reject with model-based ($p = 0.545$) or cluster-robust ($p = 0.721$) covariance. All 50 leave-one-initialization-out fits completed; all retained the negative averaged sign, 41 had raw $p < 0.05$, and estimates ranged from -0.014433 to -0.008562 . These diagnostics do not justify deletion. The frozen classification is a model-dependent directional signal, objective-specific under end-to-end mode.

Contrast or sensitivity	Estimate	Wald 95% CI	Bootstrap 95% CI
End-to-end, $L = 0$	-0.000958	[-0.015251, 0.013336]	[-0.030298, 0.024478]
End-to-end, $L = 1$	-0.022273	[-0.036493, -0.008052]	[-0.048098, 0.007609]
End-to-end, L-average	-0.011615	[-0.021697, -0.001534]	[-0.030943, 0.010259]
Active depths $D \geq 3$, L-average	-0.014788	[-0.024480, -0.005095]	Not planned
Conditional mode, L-average	0.006978	[-0.004745, 0.018701]	Not available

Table 3 Post-primary H3 explanatory and sensitivity results. Bootstrap intervals contain zero for both objective-specific contrasts and their average. Active-depth and conditional-mode rows are not substitutions for the primary model.

Dataset	Estimand	Point	Median	Percentile 95% CI
Original	$J_{EL R=0}$	1.241760	1.238674	[1.082210, 1.414864]
Original	$J_{EL R=1}$	1.126633	1.124329	[0.991331, 1.285152]
Independent seed	$J_{EL R=0}$	1.163219	1.160949	[1.007100, 1.352596]
Independent seed	$J_{EL R=1}$	1.242137	1.238157	[1.049874, 1.449258]

Table 4 Conditional descriptive exact-gradient indices. Each initialization-cluster bootstrap completed 2,000 iterations with zero failures. Values above one are above the multiplicative no-interaction reference, not percentage improvements in optimization.

6 Conditional exact-gradient indices

The frozen descriptive indices are

$$J_{EL|R=0} = \frac{G_{EL}G_0}{G_E G_L}, \quad J_{EL|R=1} = \frac{G_{ELR}G_R}{G_{ER}G_{LR}}.$$

Unique exact-gradient parameter rows are pooled before the configuration-level RMS construction, so deeper depths receive more observation/parameter weight. The indices are conditional on R , not transformations of centered coefficients.

7 Prepared-state metrics at initialization

The frozen table contains 4,000 unique terminal-block states after exact-state duplication across shot budgets was removed. Fidelity and normalized-energy bounds, row uniqueness, energy spectral bounds, and fidelity–infidelity complementarity all passed. Original configuration–depth mean fidelity ranged approximately 0.0338–0.0817 and normalized energy 0.4631–0.5529; independent-seed ranges were 0.0322–0.0962 and 0.4561–0.5537. The ranges overlap strongly. Since L changes evaluation rather than preparation, paired $L = 0/L = 1$ prepared-state summaries are identical. No significance test or causal gradient–task relationship is inferred.

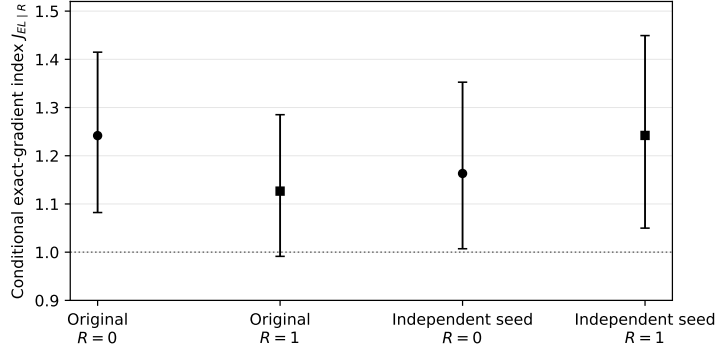


Fig. 3 All four frozen conditional $J_{EL|R}$ estimates with percentile 95% bootstrap intervals and reference line $J = 1$. Marker shape identifies R and line style identifies dataset. These are descriptive secondary ratios.

8 Measurement implementation and resource accounting

Global infidelity is sampled directly from exact statevector overlap probability as one abstract overlap setting; no inverse target-state circuit, preparation gates, or physical cost is represented. TFIM energy uses a shared Z-basis multinomial setting for three ZZ terms and a shared X-basis multinomial setting for four X terms. End-to-end mode samples $D - 1$ forward-feature Z jobs and re-encodes noisy features; conditional mode holds those features exact. Modes are not pooled.

All 320 resource rows conserve requested B exactly, allocations are deterministic and nonnegative, and no production job has zero shots. All 1,833 zero-variance cells join to complete resource records. At $D = 6$, conditional/global uses 48 jobs and end-to-end/local 61, a 27.08% increase. The experiment matches total implemented simulator shots, not circuits, jobs, shots per circuit, physical gates, calibration, readout, hardware noise, or wall-clock time.

9 Eligibility and zero-variance resource join

Exactly zero replicate variance was restricted to $L = 0$: 509 of 102,400 end-to-end and 1,324 of 102,400 conditional cells, with none under $L = 1$. The primary finite-shot estimand therefore conditions on positive estimated variance. The resource join demonstrates that these exclusions do not reflect missing accounting records. They remain an objective-linked selection limitation.

10 Historical correction and audit record

Historical direct-0/1 coefficient tables and older pooled estimator-mode outputs are retained in explicitly superseded directories for auditability. They must not be read as current primary factorial conclusions. In particular, the former H3 value near -0.000958 is the end-to-end $E \times R$ simple interaction at $L = 0$; it now appears as a

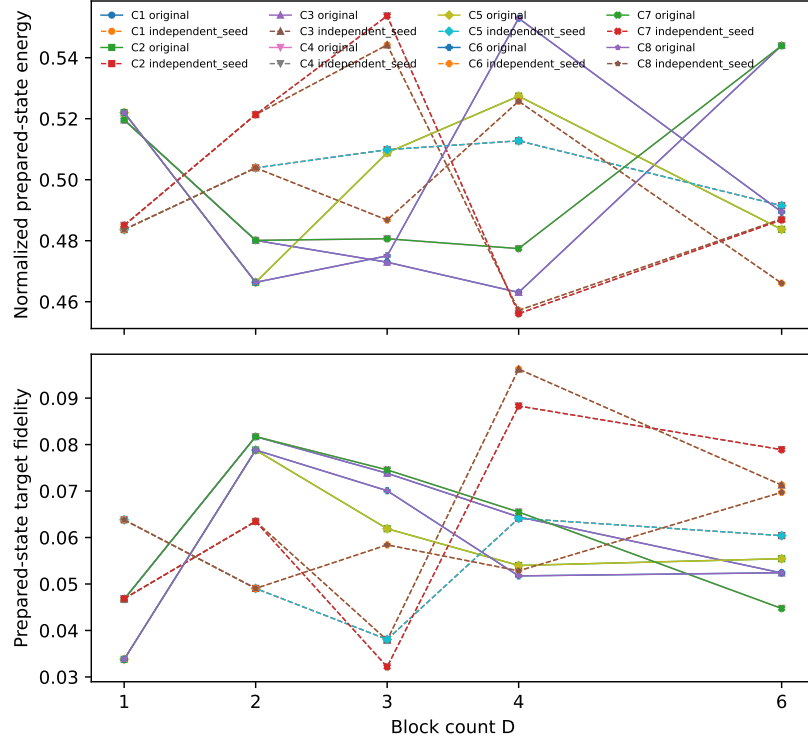


Fig. 4 Descriptive terminal-block prepared-state fidelity and normalized energy at initialization from frozen source data. No regression or optimization trajectory is implied.

post-primary explanatory contrast, not the centered H3 estimand. Historical ‘n=400’ H1 bootstrap text is superseded by 2,000 completed fits. Historical H4 Holm $p = 0.115$ is superseded by the corrected family value 0.057658.

11 Reproducibility index

12 Remaining limitations

Limitations include four qubits; one open-boundary TFIM task; reset-per-block architecture; one implementation per intervention; one initialization family; 50 top-level clusters per dataset; exact-state and abstract shot simulation; no hardware noise, read-out, calibration, or optimization trajectories; prepared-state metrics only at initialization; L -linked finite-replicate zero-variance selection; depth weighting from parameter count; covariance-sensitive depth conclusions; H2/H3 procedure disagreement; H3 estimator-mode disagreement; weighting-dependent conditional J; inaccessible full protocol text as a factor-coding provenance limitation; and no matched physical-resource claim.

Component	Frozen source
Numerical source of truth	results/final_submission_v1/manifest.json; final_numerical_results.json
Factor coding	verification/factor_coding_correction_results.md
Independent-seed H1	verification/h1_independent_seed_results.md
H1 depth/weighting	verification/h1_depth_weighting_results.md; results/h1_depth_weighting/
H3 explanation	verification/h3_centered_interaction_robustness_results.md
Conditional J	verification/jel_conditional_bootstrap_results.md; results/jel_conditional/
Prepared states	verification/task_metrics_results.md; results/task_metrics/
Resources	verification/measurement_resource_accounting.md; results/resource_accounting/
Figure 0 archive	verification/figure_source_reconciliation_results.md

Table 5 Frozen reproducibility index. Repository paths are typeset as breakable labels.